# Tharindu Ekanayake

Machine Learning Engineer

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Work authorization: EU Blue Card holder

## PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years building production ML systems, from multi-camera perception on edge hardware to multimodal RAG platforms on GPU infrastructure. Currently engineering a production RAG platform at goAVA, with responsibility across hybrid retrieval, cross-encoder reranking, GPU model serving and CI-enforced retrieval evaluation. Previously built the perception stack for an autonomous checkout system used in live retail environments. First author of an IEEE ICASSP 2026 paper on viewpoint-invariant 3D pose canonicalization.

## TECHNICAL SKILLS

**ML & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, sentence-transformers, scikit-learn

**LLM & RAG:** hybrid dense/sparse retrieval (multilingual-E5, SPLADE, RRF), cross-encoder reranking, Qdrant, LangChain, LangSmith, LLM-as-judge evaluation, OpenAI / Azure OpenAI / Anthropic / Gemini

**Computer Vision:** detection and segmentation (YOLO11, RT-DETR, SAM2), multi-object tracking and re-ID, multi-camera triangulation, 2D/3D pose estimation (ViTPose, Sapiens), LiDAR perception (OpenPCDet)

**Serving, GPU & Edge:** NVIDIA Triton, Hugging Face TEI, FastAPI, CUDA, CuPy, TensorRT, FP16, NVENC, ONNX, NVIDIA Jetson

**MLOps & Infrastructure:** Docker, Kubernetes (GKE), GitHub Actions, OpenTofu, Prometheus, Redis, PostgreSQL, HPC, Weights & Biases

**Programming:** Python, SQL, Bash

## WORK EXPERIENCE

### Machine Learning Engineer, goAVA GmbH

*Essen, Germany | 01/2026 – Present*

- Engineer a production multimodal RAG platform supporting documents, audio, video, images and tabular data, with hybrid retrieval, grounded LLM answers and per-tenant data isolation.
- Designed the retrieval pipeline using multilingual-E5 dense and SPLADE sparse retrieval fused with Reciprocal Rank Fusion, followed by BGE cross-encoder reranking over Qdrant, improving recall@5 from 0.78 to 0.90 on a 500-question golden set.
- Moved embedding and reranking from in-process PyTorch to dedicated Hugging Face TEI GPU services, raising throughput from ~0.4 to 2.6 req/s (~6.5x) with zero failures at 40 concurrent requests on a single NVIDIA L4.
- Built CI-enforced retrieval evaluation tracking recall@5, candidate recall, MRR, hit rate@5 and p95 latency, blocking merges that regress against the established baseline.
- Built distributed Q&A generation services with FastAPI and Redis Queue, with cross-document embedding deduplication, LLM-as-judge evaluation and per-customer quality contracts.
- Developed CUDA-accelerated video processing with custom CuPy kernels, FP16, NVENC and batched ViTPose / YOLO11-Pose inference for video files and RTSP streams, deployed via Triton, Docker and Kubernetes.
- Built multimodal ingestion using Docling, OCR, audio transcription and VLM image understanding, with SSE streaming, automated testing, security scanning and GitHub Actions CI/CD.

### Project Researcher, CMVS, University of Oulu

*Oulu, Finland | 10/2024 – 01/2026*

- Designed 3DPCNet, a hybrid GCN-Transformer for viewpoint-invariant 3D pose canonicalization with adaptive graph learning, cross-attention fusion and 6D rotation representation, reducing MPJPE by ~35% from 74.1 mm to 48.4 mm in cross-environment evaluation; first-author IEEE ICASSP 2026 paper.

- Adapted OpenPCDet and PV-RCNN for sparse indoor LiDAR person detection, including keypoint-derived 3D boxes, voxelization and coordinate-frame alignment, enabling a privacy-preserving HAR pipeline that reached 94.4% accuracy on 13 rehabilitation activities and 90.3% across all 27 MM-Fi activities under cross-scene, subject-independent evaluation.
- Built contactless human-sensing pipelines from monocular RGB video in VR training environments, combining 3D pose, rPPG physiological signals and temporal analysis, run on CSC Finland's Puhti and Mahti HPC clusters.

**Machine Learning Engineer, ZOOMi Technologies Inc.** *Colombo, Sri Lanka | 10/2020 – 09/2023*

- Built the perception stack for PixelMagic, an autonomous checkout system used in live retail environments, covering detection, segmentation, multi-object tracking, re-identification and pose estimation across synchronized cameras.
- Developed multi-camera tracking that maintained stable shopper identities through occlusion and viewpoint changes, using triangulation across overlapping views.
- Optimized computer vision models for NVIDIA Jetson using TensorRT, increasing inference performance from ~12–15 FPS to ~28–30 FPS while operating within edge memory and compute constraints.
- Engineered real-time multiprocess video pipelines that kept live camera streams within frame budgets and handled failure cases encountered in real retail environments.
- Contributed to LayerNext, a computer vision data and training platform covering video ingestion, annotation and dataset versioning, and delivered ML systems for medical imaging and autonomous agricultural navigation.

## PROJECTS

**PhysioTrack, Contactless Human-Sensing Library** *Open source | 05/2025 – Present*

- Maintain a pip-installable GPL-3.0 Python library (github.com/tharindu326/physiotrack) unifying detection, tracking, 2D/3D pose, segmentation, monocular depth and rPPG behind a single prediction API.
- Integrated 51 pretrained model variants with on-demand Hugging Face weight resolution and caching for Docker deployments; built MkDocs documentation, pytest coverage and semantic-release CI via GitHub Actions.

## EDUCATION

**M.Sc. Computer Science and Engineering, Artificial Intelligence** *Finland | 09/2023 – 05/2025*

University of Oulu

*Thesis:* Unsupervised multimodal biosignal analysis of student cognitive and physiological responses during VR-based training sessions | *Grade:* 5/5 (highest)

**B.Sc. Electrical and Electronic Engineering** *Sri Lanka | 2016 – 04/2020*

University of Peradeniya

## PUBLICATIONS

- **T. Ekanayake**, C. Á. Casado, M. B. López. 3DPCNet: Pose Canonicalization for Robust Viewpoint-Invariant 3D Kinematic Analysis from Monocular RGB Cameras. *IEEE ICASSP*, 2026. First author. DOI: 10.1109/ICASSP55912.2026.11464754
- S. Sharifipour, C. Á. Casado, L. Nguyen, **T. Ekanayake**, M. L. Cañellas, et al. LiDAR-Based Human Activity Recognition through Laplacian Spectral Analysis. *IEEE ICASSP*, 2026. Co-author. DOI: 10.1109/ICASSP55912.2026.11462769
- **T. Ekanayake**, C. Á. Casado, N. Nguyen, M. Sobocinski, et al. Evaluating the Accuracy and Reliability of Camera-Based Physiological and Motion Signal Extraction Techniques in Virtual Reality Training Environments. *SCIA*, *Springer LNCS*, 2025. First author. DOI: 10.1007/978-3-031-95911-0_31